

Tutorato di Algoritmi e Laboratorio

Lezione 3 20/12/2021

Esercizi Base

1) $m^3 + 5m + 2$ $\stackrel{?}{\in} O(5m^3 + 2m + 7)$ $\checkmark$ $\begin{matrix} b \leq r \\ 3 \leq 3 \end{matrix}$ $\checkmark$ polinomi

2) $m^2 + 7$ $\stackrel{?}{\in} O(\sqrt{m^5})$ vero $\lim_{n \rightarrow \infty} \frac{m^3 + 5m + 2}{5m^3 + 2m + 7} = 0.4$

3) $2^m + 7m^2$ $\stackrel{?}{\in} O(m^{10} + 7m^3)$ falso

4) $4m^3$ $\stackrel{?}{\in} \Omega(m^2 + 8)$ $3 \geq 2$ vero

5) $4m^3$ $\stackrel{?}{\in} \Omega(\cancel{7m^2} + \sqrt{m^7 + 2m})$ $\frac{3}{2} = 3.5$ $\frac{3}{2} \geq 3.5$ falso

6) $2^m + 7m$ $\stackrel{?}{\in} \Omega(3m^3)$ vero

7) $2^m + 2m$ $\stackrel{?}{\in} \Omega(\cancel{2^m} + 3m^3)$ vero

8) $3m^4 + 2$ $\stackrel{?}{\in} \Theta(m^4)$ $1 \geq 1 \wedge 1 \leq 1$ $1 = 1$ vero

9) $2m^3 + 3m^4$ $\stackrel{?}{\in} \Theta(\sqrt{m})$ $m^{\frac{1}{2}}$ falso

10) $2m^{3.5} + 7$ $\stackrel{?}{\in} \Theta(2\sqrt{m^5})$ $\frac{5}{2} = 2.5$ vero

11) $3m^4$ $\stackrel{?}{\in} o(m^2 + 3m)$ $4 < 2$ vero

12) $2m^2$ $\stackrel{?}{\in} \omega(2m + 2)$ $2 > 1$ vero

Svolgimento...

$$\underline{m^4} + 4m^2 + 2 \stackrel{?}{\in} O(\underline{m^5} + 2m^3) \quad \checkmark$$

$4 \leq 5 \quad \checkmark$

$$m^7 + 2m^2 \stackrel{?}{\in} \Omega(m^6 + 2)$$

$7 \geq 6$

$$m^7 + 2 \stackrel{?}{\in} \Theta(m^6) \quad \times \text{ non appartiene}$$

sia che $7 \leq 6$ che $7 \geq 6$
falso and true falso

$$m^2 + 7 \stackrel{?}{\in} O(\sqrt{m^5}) \quad \text{vera}$$

$$m^2 + 7 \stackrel{?}{\in} O(m^{\frac{5}{2}})$$

$$2 \leq \frac{5}{2} = 2,5$$

$$\underline{2^m} + 5m^{1000} \stackrel{?}{\in} O(m^{1000} + 2^{m+1}) \quad \text{vero}$$

$m^{1000} + 2 \cdot 2^m$

$$z^n + 2n \stackrel{?}{\in} \Omega(z^n + 3n^3) \quad \text{vera!}$$

$$\exists n_0 > 0 : \exists c : \forall n > n_0 \quad z^n + 2n \geq c(z^n + 3n^3)$$

$$z^n + 2n \geq cz^n + c3n^3$$

$$z^n - cz^n \geq c3n^3 - 2n$$

$$\underline{(1-c)z^n} \geq \underline{3cn^3 - 2n}$$

↓

esponenziale

↓

polinomio

vero per $n > n_0$

Corollario del Teorema Masters (importante!)

$$T(n) = \underline{a} T(\underline{\frac{n}{b}}) + \Theta(n^k \log_2^h n)$$

Condizioni su a e su b ?

$$T(n) = \begin{cases} \Theta(n^{\log_b a}) & \text{se? } \log_b a > k \\ \Theta(n^k \log_2^{h+1} n) & \text{se? } \log_b a = k \\ \Theta(n^k \log_2^h n) & \text{se? } 0 < \log_b a < k \end{cases}$$

$$T(n) = \underline{a} T(\underline{\frac{n}{b}}) + \Theta(n^k \log_2^h n)$$

per risolvere il problema di dimensione n

→ applichiamo a volte $\Leftrightarrow a \geq 1$

→ il problema di dimensione $b > 1$

$n < \frac{n}{b}$ se $b > 1$

$$\underset{\uparrow}{T(n)} = \underline{a} \underset{\uparrow}{T(n)} + \Theta(n^k \log_2^h n)$$

Esercizio (prova del 7 Dicembre 2021)

$$T(n) = a T\left(\frac{n}{3}\right) + \Theta(n^3 \log^2 n)$$

1) risolvere

2) quando $T(n) = \Theta(n^4)$

3) quando $T(n) = \Omega(n^3 \log^4 n)$

4) quando $T(n) = o(n^3 \log^3 n)$

$$T(n) = a T\left(\frac{n}{3}\right) + \Theta(n^3 \log^2 n)$$

$$a = a$$

$$b = 3$$

$$\log_b a \stackrel{?}{\geq} K$$

$$K = 3$$

$$\log_3 a \stackrel{?}{\geq} 3$$

$$h = 2$$

$$\log_3 a < 3 \cdot 1$$

$$1 = \log_3 3^1$$

$$\log_3 a < \log_3 3^3$$

$$\log_3 a = 3$$

$$\log_3 a = \log_3 3^3$$

$$\log_3 a > 3$$

$$\log_3 a < \log_3 27$$

$$a = 27$$

$$a > 27$$

$$a < 27$$

$$1 \leq a < 27$$

$$\Theta(n^3 \log_2^2 n)$$

$$\Theta(n^3 \log_2^3 n)$$

$$\Theta(n^{\log_b a})$$

2) $\Theta(n^3 \log_2^2 n) \stackrel{?}{=} \Theta(n^4)$

falso tra 1 e 27

$\rightarrow n^3 \log_2^3 n \stackrel{?}{=} \Theta(n^4)$ falso per $n = 27$

$\rightarrow n^{\log_b a} \stackrel{?}{=} \Theta(n^4)$

$$\log_b a = 4 \quad \Leftrightarrow \log_3 a = 4 \cdot 1$$

$$\log_3 a = 4 \log_3 3$$

$$\log_3 a = \log_3 3^4 = \log_3 81$$

$$a = 81$$

$$1 < a < 27$$

$$\Theta(n^3 \log_2^2 n)$$

$$a = 27$$

$$\Theta(n^3 \log_2^3 n)$$

$$a > 27$$

$$\Theta(n^{\log_b a})$$

$$T(n) = \Omega(n^3 \log^4 n)$$

$$\rightarrow n^{\textcircled{3}} \log_2^{\textcircled{2}} n \stackrel{?}{\in} \Omega(n^{\textcircled{3}} \log^{\textcircled{4}} n)$$

$$2 \geq 4 \quad \text{falso!}$$

$$\rightarrow n^{\textcircled{3}} \log_2^{\textcircled{3}} n \stackrel{?}{\in} \Omega(n^{\textcircled{3}} \log^{\textcircled{4}} n)$$

$$3 \geq 4 \quad \text{falso!}$$

$$\rightarrow n^{\textcircled{\log_b a}} \stackrel{?}{\in} \Omega(n^{\textcircled{3}} \log^4 n)$$

$$\log_3 a \geq 3.1$$

$$\log_3 a \geq \log_3 3^3$$

$$\begin{cases} a \geq 27 \\ a > 27 \end{cases} \quad \Rightarrow \quad a > 27$$

siano nel 3° caso!

$$1 < n < 27$$

$$\Theta(n^3 \log_2^2 n)$$

$$n = 27$$

$$\Theta(n^3 \log_2^3 n)$$

$$n > 27$$

$$\Theta(n^{\log_2 3})$$

$$T(n) = O(n^3 \log^3 n)$$

$$n^{\textcircled{3}} \log_2^{\textcircled{2}} n \stackrel{?}{\in} O(n^{\textcircled{3}} \log^{\textcircled{3}} n)$$

$$2 < 3 \quad \text{vero} \quad \text{per } 1 < a < 27$$

$$\hookrightarrow n^{\textcircled{3}} \log_2^{\textcircled{3}} n \stackrel{?}{\in} O(n^{\textcircled{3}} \log^{\textcircled{3}} n)$$

$$3 < 3 \quad \text{falso!}$$

$$n^{\log_3 a} \stackrel{?}{\in} O(n^{\textcircled{3}} \log^3 n)$$

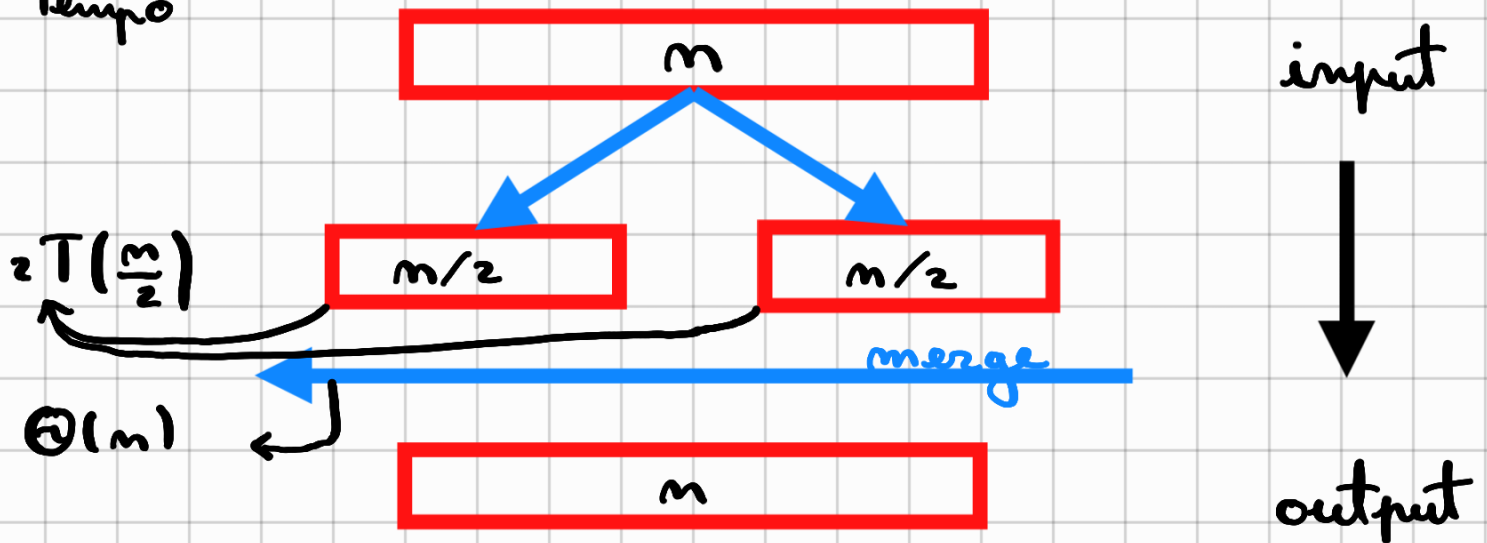
$$\log_3 a < 3.1$$

$$\begin{cases} a < 27 \\ a > 27 \end{cases} \Rightarrow \emptyset$$

Esercizio

- Qual'è il tempo di esecuzione di mergesort?
- Come ci aiuta il Teorema Masters?

Tempo



$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta\left(n^1 + \underbrace{\log_2 n}_1\right)$$

$$a = 2$$

$$b = 2$$

$$\log_2 2 = 1 = 1 = k$$

$$T(n) = \Theta(n \log_2 n)$$

Codici di Huffman

"Tutorato per tutti:"

	"frequenza" #occorrenze	lunghezza binaria	Huffman
" "	2	0	0000
a	1	1	0001
e	1	2	0010
i	1	3	0011
o	2	4	0100
t	1	5	0101
r	2	6	0110
e	6	7	0111
u	2	8	1000
TOT	18		

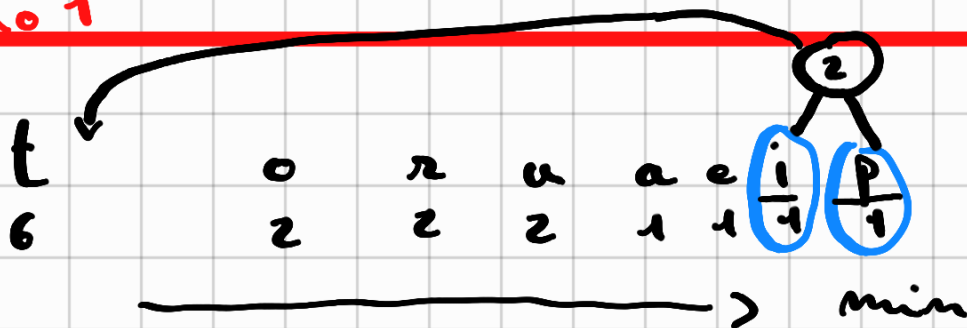
n "numero di caratteri differenti" = 9

$$\# \text{bit} = \lceil \log_2 9 \rceil$$

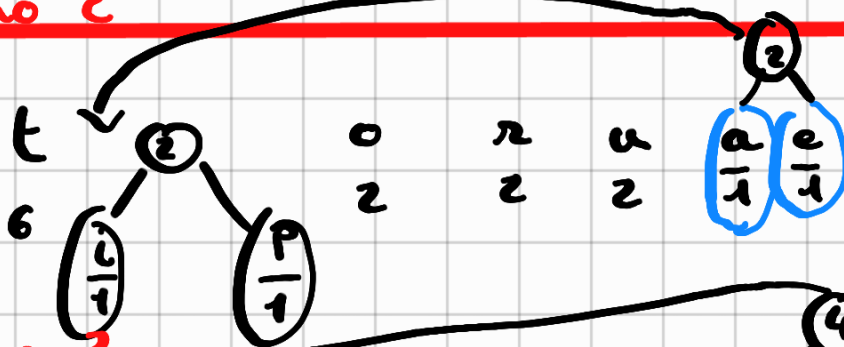
$$a \frac{1}{2^0} + b \frac{2}{2^1} + c \frac{4}{2^2} + d \frac{8}{2^3}$$

I calcoli seguenti e' come se non
ci fosse lo spazio

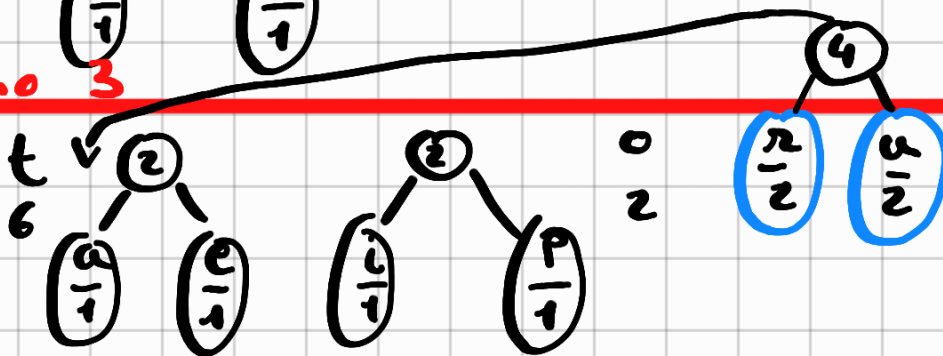
Tempo 1



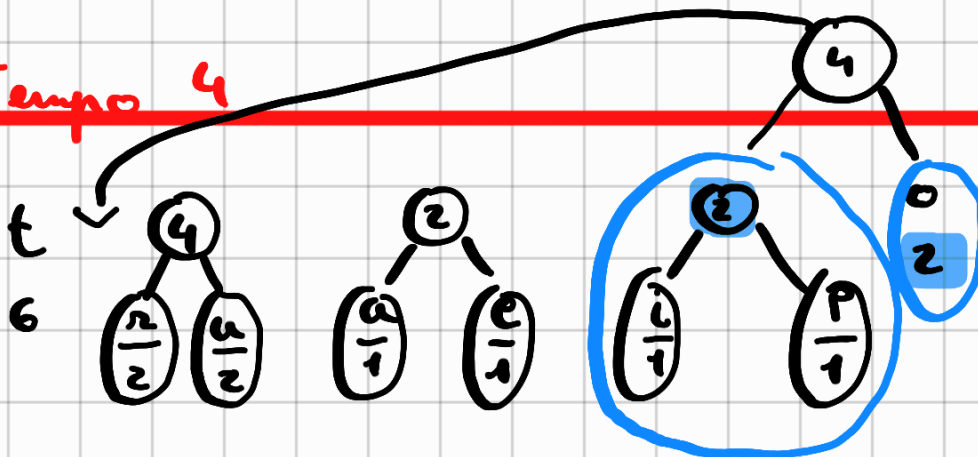
Tempo 2



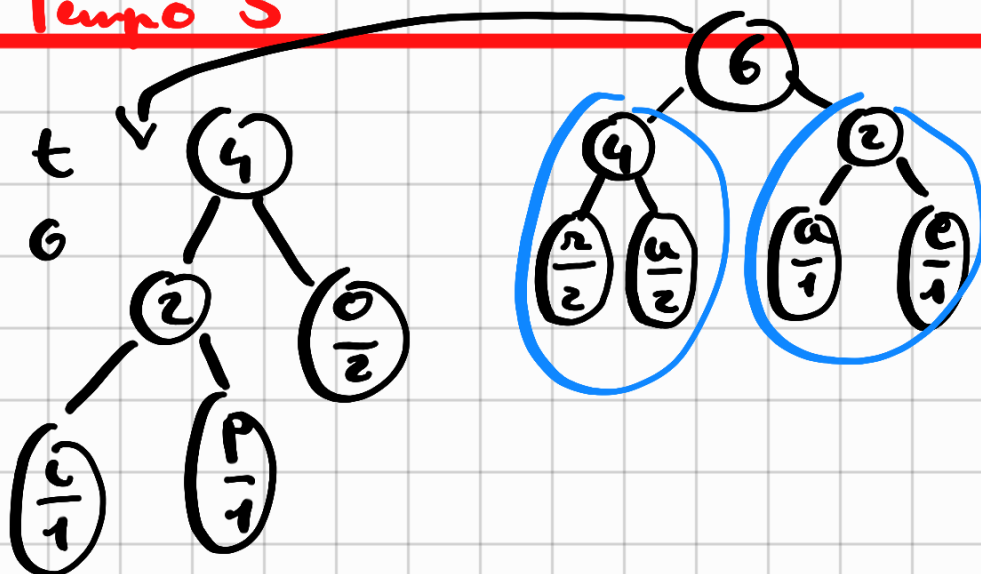
Tempo 3



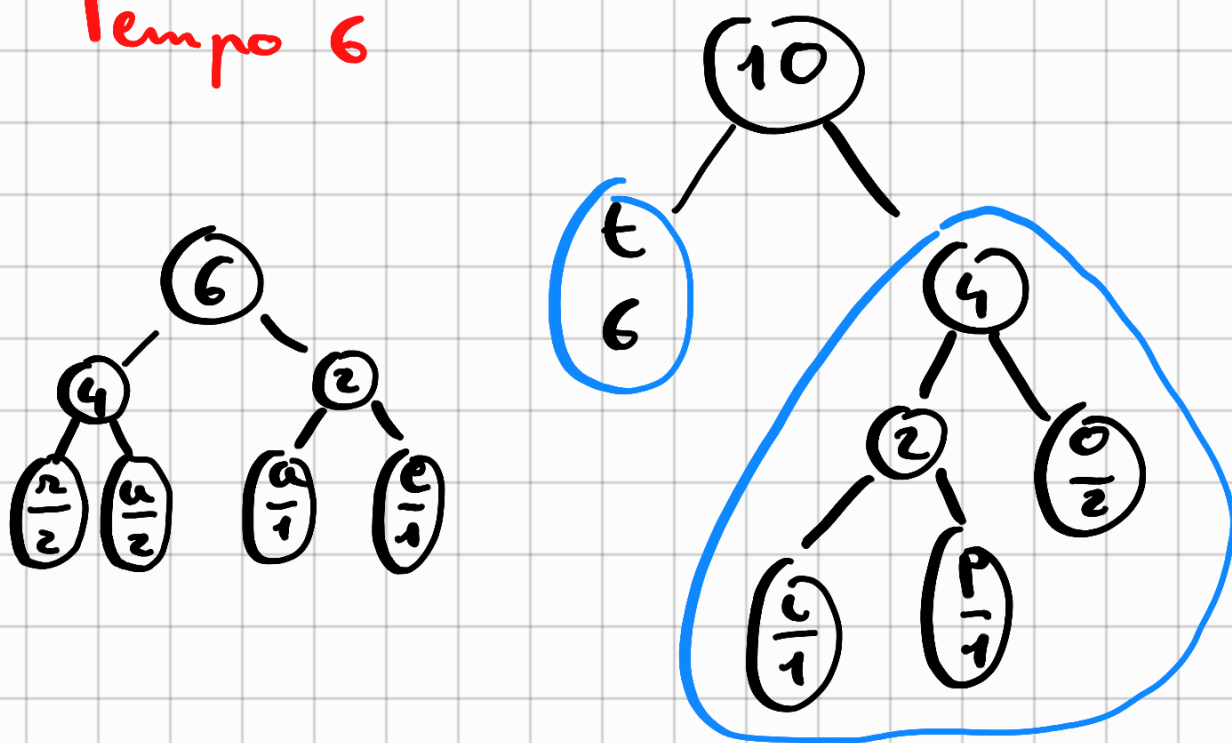
Tempo 4



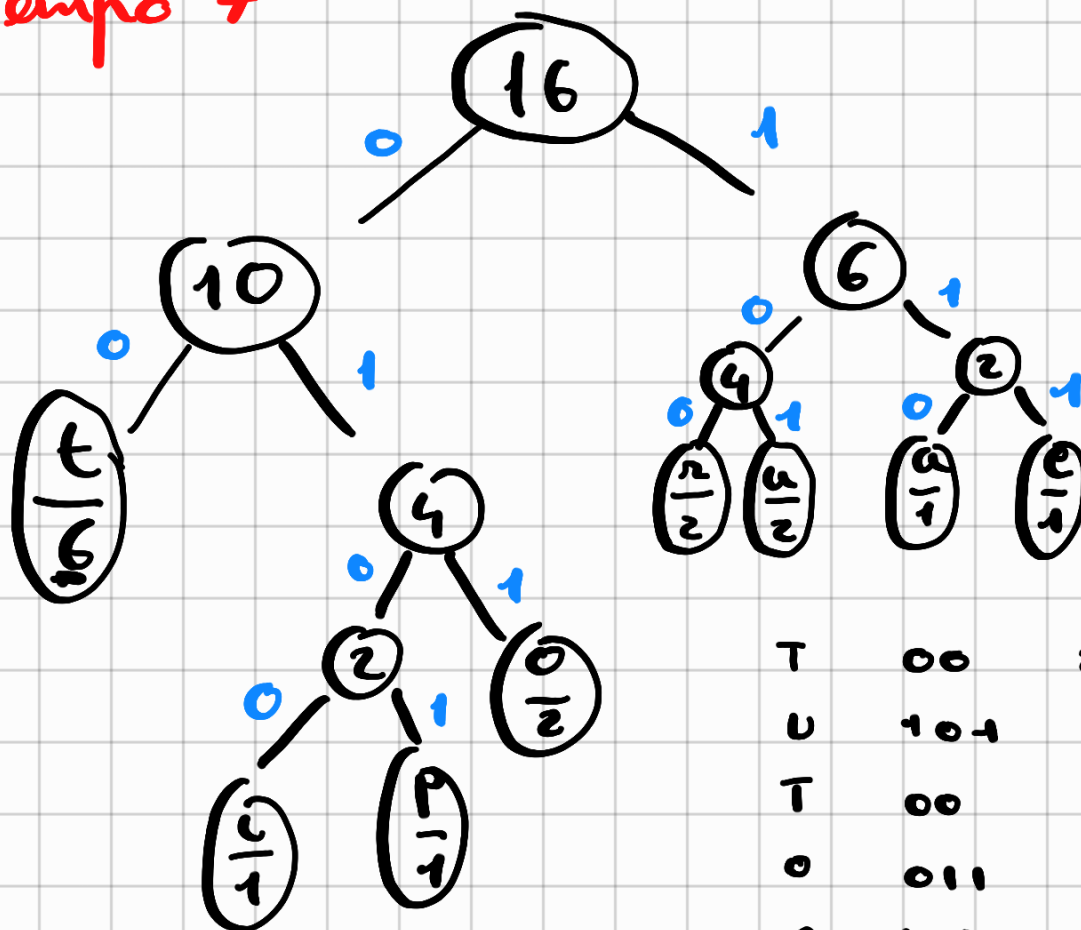
Tempo 5



Tempo 6



Tempo 7



T	00	2 bit	12 bit
U	101		6 bit
T	00		
0	011		
2	100		